Matrices & Systems

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

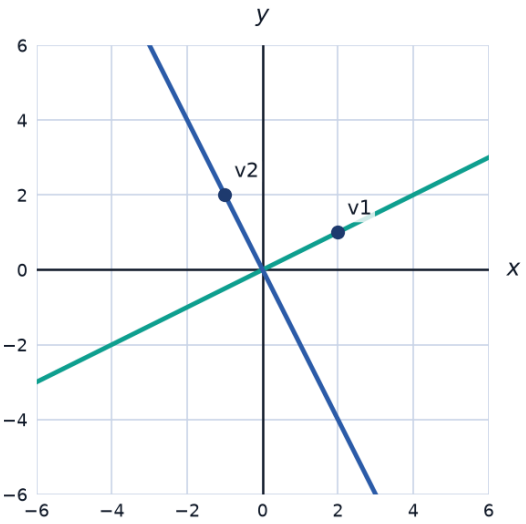
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [4, -2; 1, 3]$ and $B = [-1, 5; 2, 0]$, what is the matrix sum $A + B$?

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = (3, -2; 1, 4)$ and $B = (-5, 2; 0, -3)$, what is the matrix $2A - B$?



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the figure, right triangle ABC has a right angle at vertex C, and altitude CD is drawn perpendicular to hypotenuse AB. If $AD = x$, $DB = x + 5$, and $CD = 6$, what is the length of hypotenuse AB?

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

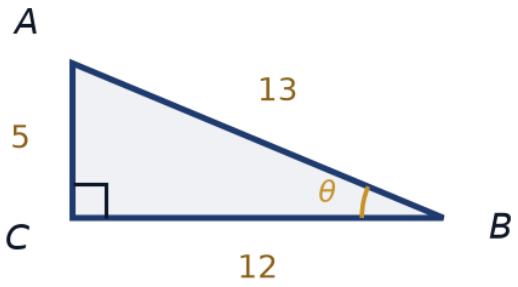
5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

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6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a right-angled triangle, one of the acute angles measures 30°. If the side opposite this angle has a length of 5 units, what is the length of the hypotenuse?



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = (5, -1; 2, 4)$ and $B = (k, 2; -2, 5)$, for what value of the constant k is the matrix $C = 2A - B$ not invertible?

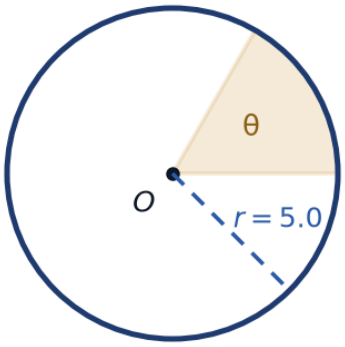
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

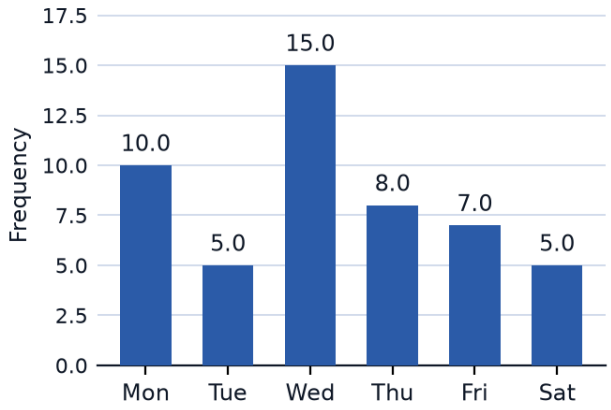
What are the coordinates of the center and the radius of the circle given by the equation $(x - 4)^2 + (y + 1)^2 = 36$?



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10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A small bakery recorded the number of muffins sold each day for a week. The results are shown in the bar chart below. What is the total number of muffins sold during the week?



11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $(x - 2)$. When $P(x)$ is divided by $(x + 1)$, the remainder is -12. What is the remainder when $P(x)$ is divided by $(x - 3)$?

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An ellipse in the xy-plane is defined by the equation $9x^2 + 16y^2 - 54x + 64y + c = 0$ where c is a constant. If the length of the minor axis of the ellipse is 6, what is the value of c?

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The polynomial $P(x) = x^4 + ax^3 - 7x^2 + bx - 12$ has zeroes at $x = -3$ and $x = 2$, where a and b are real constants. What is the value of $a + b$?

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

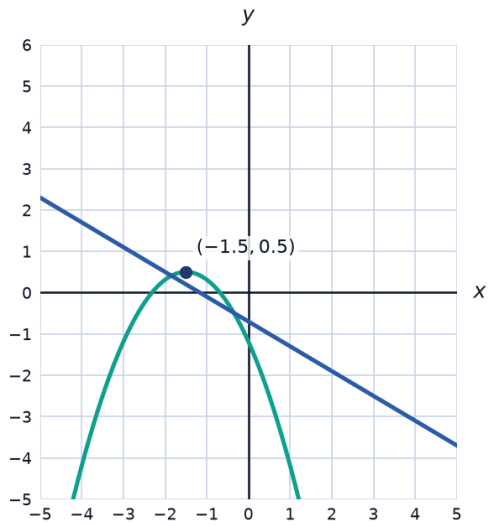
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A hyperbola is defined by the equation $\frac{(x-h)^2}{16} - \frac{(y-k)^2}{9} = 1$. If the vertices of the hyperbola are located at $(5, 2)$ and $(-1, 2)$, determine the values of h and k.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Radicals & Rational Exponents

For all real values of x such that $xeq-1$ and $xeq2$, the equation $\frac{ax+7}{x+1}-\frac{b}{x-2}=\frac{2x^2-3x-20}{x^2-x-2}$ is true, where a and b are constants. What is the value of $a+b$?



17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

A manufacturing process produces items, and the number of defects per batch of 100 items follows a Poisson distribution with a mean of λ. The cost of inspecting a batch is C_i dollars, and the cost of repairing a defective item is C_r dollars. If a batch has k defects, the total cost for that batch is $C(k)=C_i+k\cdot C_r$. If $\lambda=3$, $C_i=50$, and $C_r=10$, find the expected total cost per batch, given that the expected number of defects in a batch is $E[K]=\lambda$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

Consider the equation $\sqrt{2x+k}=x-1$, where k is a constant. If $x=5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions

A set of exam scores is normally distributed with a mean of $\mu=74$ and a standard deviation of $\sigma=8$. What is the z-score corresponding to an exam score of 58?

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions & Their Graphs

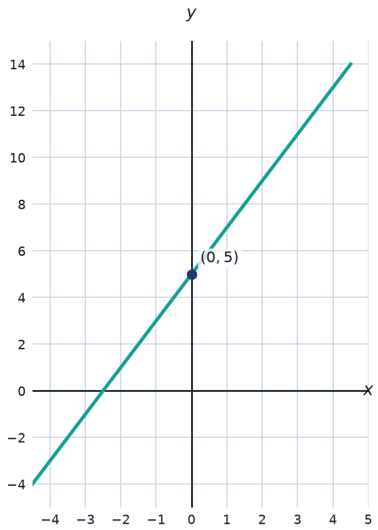
Let $f(x)=\sqrt{x^2-6x+9}$ and $g(x)=x-3$. For which values of x is $f(x)=g(x)$?

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(2)$.



a. 9

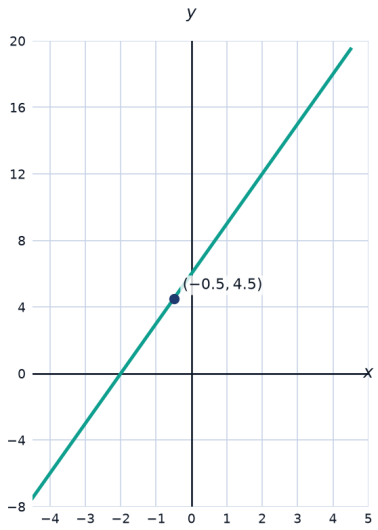
b. 11

c. 4

d. 13

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(2)$.



a. 12

b. 15

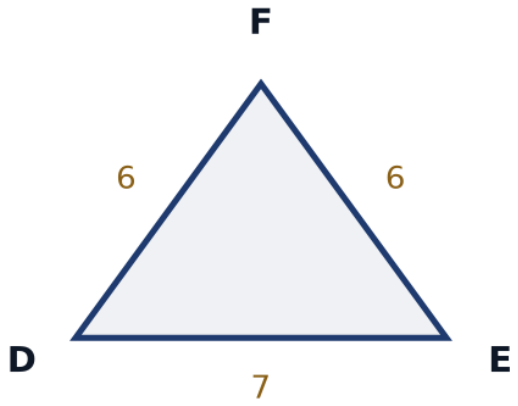
c. 6

d. 16

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angle sum and difference identities, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 88°

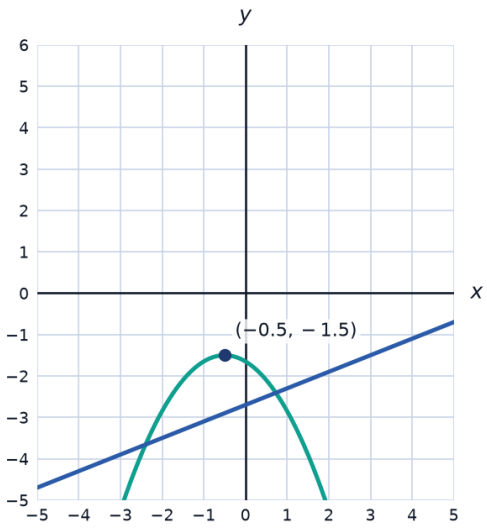
b. 98°

c. 80°

d. 133°

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

c. 24 cm

d. 54 cm

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for reciprocal, quotient, and pythaogrean identities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cofunction identities consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12

d. 17

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for verifying trig identities consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 8$ in verifying trig identities, evaluate $f(3)$.

a. 23

b. 28

c. 15

d. 27

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For eigenvalue λ with eigenvector v, any scalar multiple cv is also an eigenvector. True or False?

a. Only if $c = 1$

b. False

c. Only if $c > 0$

d. True

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the matrix equation $(2, 1, -2; 4, -1, -1; 2, -2, 1)(x; y; z) = (9; 9; k)$ where k is a real constant. Given that the system has infinitely many real solutions (x, y, z), what is the minimum possible value of $x^2 + y^2 + z^2$?

a. 9

b. 3

c. 6

d. 45

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the system of linear equations represented by the matrix equation $(1, 1, 1; 1, 2, 3; 3, 5, k)(x; y; z) = (6; 14; m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x, y, z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2 = x^2 + y^2 + z^2$. What is the value of $k + m + d^2$?

a. 55

b. 47

c. 38

d. 109

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $A = (x, 2; 3, x - 1)$. If A is not invertible and $x > 0$, what is the value of $\det(A + 3I)$, where I is the 2×2 identity matrix?

a. 24

b. 9

c. 0

d. -6

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34 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the matrix equation: $(3, -1; 2, k)(x; y) = (5; 7)$. If the matrix $(3, -1; 2, k)$ is singular, what is the value of k?

a. $-3/2$

b. $3/2$

c. $-2/3$

d. $2/3$

35 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Given the matrix $A = (2, 1; 3, 4)$ and the matrix $B = (1, -1; 0, 2)$, what is the resulting matrix $A + B$?

a. $(3, 0; 3, 6)$

b. $(2, -1; 3, 2)$

c. $(1, 2; 3, 2)$

d. $(3, 1; 3, 4)$

36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Given the matrix $A = (2, 1; 3, 4)$ and the matrix $B = (1, 0; 0, 1)$, what is the resulting matrix when A is added to B?

a. $(2, 1; 3, 4)$

b. $(3, 1; 3, 5)$

c. $(1, 1; 3, 3)$

d. $(2, 0; 0, 4)$

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Prove that the determinant of a matrix with two identical rows is zero.

a. Only for 2×2

b. Only if symmetric

c. Swapping the two identical rows changes the sign of the det but doesn't change the matrix: $\det = -\det \Rightarrow \det = 0$. ✓

d. Not true

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Conic Sections

State the rank-factorization theorem.

a. $A = A^T A$

b. Only for square matrices

c. Not always possible

d. Any $m \times n$ matrix A of rank r can be written as $A = CR$ where C is $m \times r$ and R is $r \times n$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Trigonometric Functions

State the Sylvester rank inequality: for $A_{m \times n}$ and $B_{n \times p}$:

a. $\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$

b. $\text{rank}(AB) = \text{rank}(A) + \text{rank}(B)$

c. $\text{rank}(A) + \text{rank}(B) - n \leq \text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$

d. $\text{rank}(AB) = 0$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Functions & Their Graphs

Prove that $\text{rank}(A) + \text{nullity}(A) = n$ (rank-nullity theorem).

a. By the fundamental theorem of linear algebra: column space dim + null space dim = number of columns

b. Not true

c. Only for 2×2

d. Only for square matrices

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability & Statistics

Prove: for invertible A, the eigenvalues of A^{-1} are $1/\lambda_i$.

a. If $Av = \lambda v$ then $v = \lambda A^{-1}v \Rightarrow A^{-1}v = (1/\lambda)v$. ✓

b. Only for diagonal matrices

c. Only if $\lambda > 0$

d. Not true

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Show that if A has eigenvalue λ, then A^n has eigenvalue λ^n.

a. Not always true

b. Only for symmetric A

c. Use Cayley-Hamilton

d. $A^n v = A^{n-1}(Av) = A^{n-1}(\lambda v) = \lambda A^{n-1}v = \cdots = \lambda^n v$. ✓

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Polynomial Functions

State the Cayley-Hamilton theorem.

a. Only 2×2 matrices

b. Only diagonalizable matrices satisfy it

c. Every square matrix satisfies its own characteristic equation

d. It applies to vectors

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rational Expressions

Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for any $m \times n$ matrices A and $n \times m$ matrix B.

a. Only if $A = B$

b. Not true in general

c. $\operatorname{tr}(AB) = \sum_{i,j} a_{ij} b_{ji} = \operatorname{tr}(BA)$. ✓

d. Only for square matrices

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the recurrence $(x_{n+1}; y_{n+1}) = A(x_n; y_n)$, the solution is:

a. $(x_n; y_n) = nA(x_0; y_0)$

b. $(x_n; y_n) = e^{An}(x_0; y_0)$

c. Need eigenvalues first

d. $(x_n; y_n) = A^n(x_0; y_0)$

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